Week 9 — Activity: Thermal vs Chemical Diffusion

Geophysics 3A: Potential Fields & Geothermics

May 1, 2026

Purpose

To compare transient thermal diffusion with chemical diffusion processes familiar from igneous and metamorphic petrology.

Diffusion Framework

Both thermal and chemical diffusion can be written in the general form

$$\frac{\partial \phi}{\partial t} = D\nabla^2\phi,$$

where ϕ represents temperature or chemical concentration, and D is a diffusivity.

Tasks

1. Explain why both thermal and chemical diffusion act to smooth spatial gradients over time.
2. Chemical diffusion in minerals often preserves sharp zoning. Give two reasons why sharp compositional gradients may survive despite diffusion.
3. Describe how multi-component diffusion differs from thermal diffusion involving a single scalar field.

Comparison

- Thermal diffusion: single field, relatively uniform diffusivity.
- Chemical diffusion: multi-component, potentially rate-limited by the slowest species.

Reflection

Why is heat generally a poorer long-term recorder of geological processes than composition?